

Chapter 15

Complexation and Precipitation Reactions and Titrations
Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 15 Problems:

15-1, 15-17, 15-18, 15-19, 15-20, 15-21, 15-27, 15-29

Complexation reactions are widely used in analytical chemistry. Some complexes are sparingly soluble and can be used in gravimetric analysis.

The most useful complexing reagents are organic compounds that contain several electron-donor groups that can form multiple covalent bonds with metal ions.

Inorganic complexing agents are also used to control solubility, form colored species, or form precipitates.

The Formation of Complexes

Metal ions react with electron-pair donors (Lewis Bases) to form coordinate compounds or complexes.

A ligand is an ion or molecule that forms a covalent bond with a cation or neutral metal atom by donating a pair of electrons, which are then shared by the two

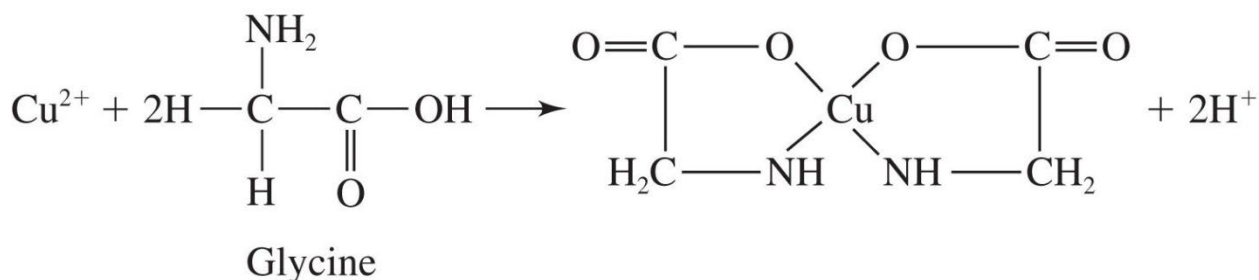
Inorganic Ligands (H_2O , NH_3 , halides)

The number of covalent bonds that a cation tends to form with electron donors is its coordination number. Typically 2, 4, or 6

Complexometric Titrations are based on complex formation.

They are based on group of coordination compounds called chelates (derived from the Greek word for claw).

A chelate is produced when a metal ion coordinates with two or more donor groups of a single ligand to form a five or six membered heterocyclic ring.



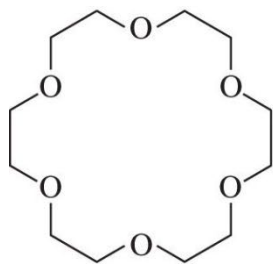
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A ligand that has a single donor group is called unidentate (NH_3 , H_2O , OH^- , Cl^-)

Dentate comes from the Latin and means having tooth like projections.

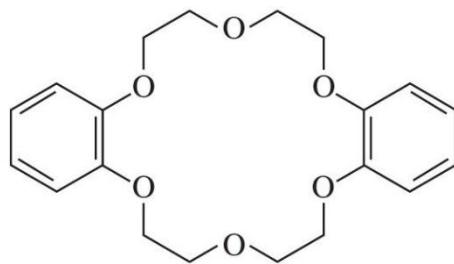
Ligands that have two donor sites are called bidentate (glycine, ethylenediamine)

Macrocycles are complexes that form between metal ions and cyclic organic compounds. These ligands contain nine or more atoms and include at least three heteroatoms, usually nitrogen, sulfur, or oxygen.

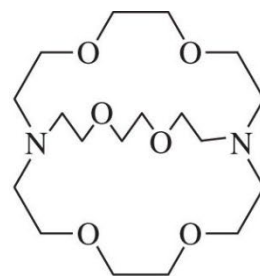


18-crown-6

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dibenzo-18-crown-6

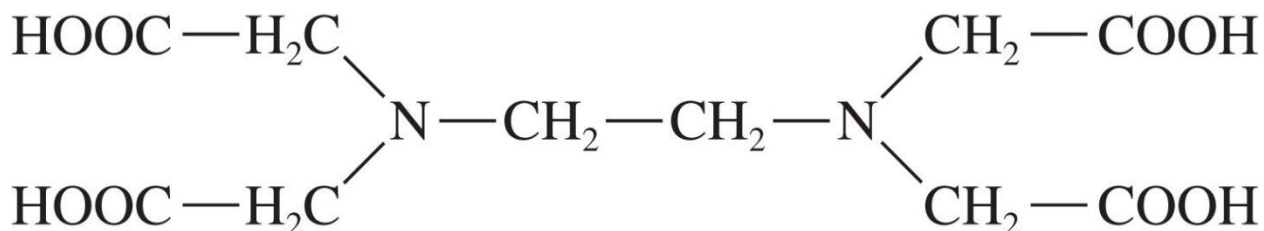


cryptand 2,2,2

Aminocarboxylic Acid Titrations

Ethylenediaminetetraacetic Acid (EDTA)

Most widely used complexometric titrant



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EDTA has six potential binding sites (the four carboxyl groups and the two amino groups) making it a hexadentate ligand

$$K_{a1} = 1.02 \times 10^{-2}$$

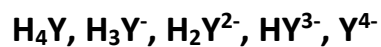
$$K_{a2} = 2.14 \times 10^{-3}$$

$$K_{a3} = 6.92 \times 10^{-7}$$

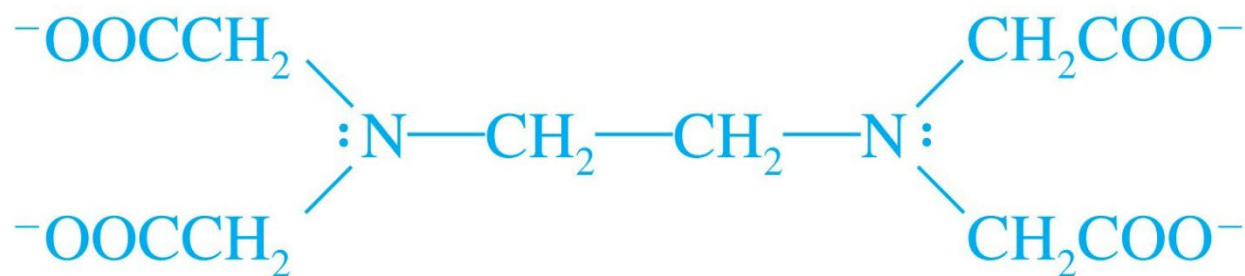
$$K_{a4} = 5.50 \times 10^{-11}$$

First two protons come off easily from opposite ends, the third and fourth protons are harder to remove (*that's why we made a basic buffer solution for the EDTA in lab*)

The various species of EDTA are often abbreviated



1^o Standard made in lab by dissolving an exactly known amount of Na₂EDTA dehydrate)



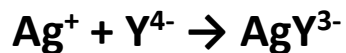
(e) Y^{4-}

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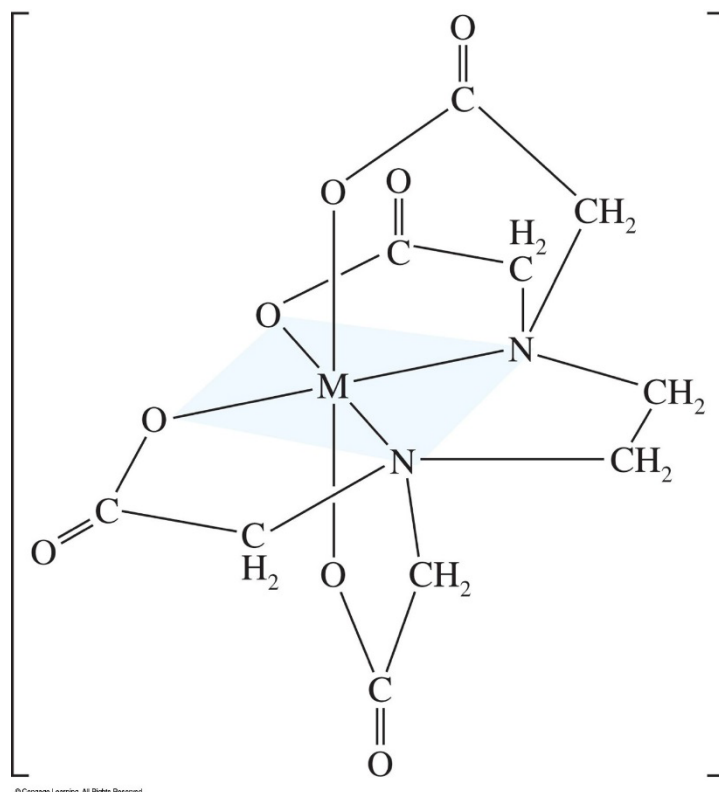
EDTA is a tetrafunctional weak acid hexadentate ligand

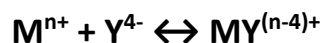
Complexes of EDTA and Metal Ions

EDTA combines with metals ions in a 1:1 ratio regardless of the charge on the cation



EDTA is remarkable not only because it forms chelates with all cations but also because most of the chelates are sufficiently stable for titrations. The stability results from the several complexing sites with the molecule that give rise to a cage like structure





K_{MY} = Formation Constant (see pg. 383 in the text)

$$k_{MY} = \frac{[MY^{(n-4)+}]}{[M^{n+}][Y^{4-}]}$$

The equation above cannot be applied if pH is variable, therefore titrations are carried out in a buffer of known pH.

$$C_T = [Y^{4-}] + [HY^{3-}] + [H_2Y^{2-}] + [H_3Y^{-}] + [H_4Y]$$

$$\alpha_4 = \frac{[Y^{4-}]}{C_T}$$

$$[Y^{4-}] = \alpha_4 C_T$$

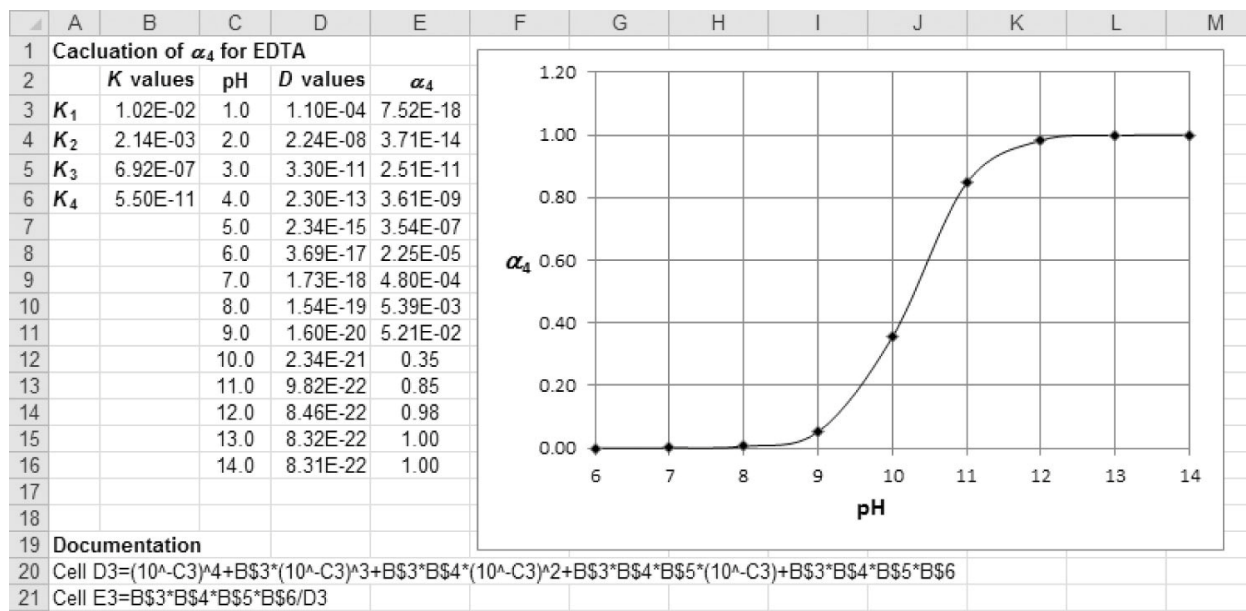
Conditional Formation Constants (k'_{MY})

$$k_{MY} = \frac{[MY^{(n-4)+}]}{[M^{n+}]\alpha_4 C_T}$$

$$k'_{MY} = \alpha_4 k_{MY}$$

CONSTANT ONLY AT THE pH α_4 APPLIES

$$k'_{MY} = \alpha_4 k_{MY} = \frac{[MY^{(n-4)+}]}{[M^{n+}]C_T}$$



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$$\alpha_4 = \frac{k_{a1} k_{a2} k_{a3} k_{a4}}{[H_3O^+]^4 + k_{a1}[H_3O^+]^3 + k_{a1}k_{a2}[H_3O^+]^2 + k_{a1} k_{a2}k_{a3}[H_3O^+] + k_{a1} k_{a2}k_{a3}k_{a4}}$$

$$\alpha_4 = \frac{k_{a1} k_{a2} k_{a3} k_{a4}}{D}$$

D= denominator

See D values in the Figure above

1. Calculate the molar Y^{4-} concentration in a 0.0300 M EDTA solution buffered to pH = 10.00.

2. Calculate the $[\text{Ni}^{2+}]$ in a solution with analytical concentration of NiY^{2-} of 0.0150 M at a pH of 8.0.

3. Calculate the $[\text{Ni}^{2+}]$ in a solution that was prepared by mixing 50.0 mL of 0.0300 M Ni^{2+} with 50.0 mL of 0.0500 M EDTA. The solution is buffered to a pH of 3.00.